What Near-Perfect Accuracy Does and Does Not Show on the UCI TCGA Pan-Cancer RNA-Seq Dataset: Leakage, Equivalence, Conformal Reliability and Data Efficiency

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Track 2: Genomics, Multi-Omics & Precision Health

Abstract

The UCI 'gene expression cancer RNA-Seq' dataset (801 TCGA tumours, 20,531 anonymised genes, five cancer types) is widely used to report near-perfect tumour-type classification, often with gene selection fitted on the full data, single splits and no statistical comparison. We asked how much these practices matter here and what the accuracies can support. With selection inside every training fold and 10×5 repeated cross-validation, seven classifiers reached 98.9–99.8% macro-F1. With real labels and all 801 samples, leaky selection changed accuracy by at most 0.29 point for 20 or more genes; on permuted labels it inflated accuracy by 65.1 ± 4.8 points at n = 30 and 14.1 ± 1.7 at n = 390 (mean, 95% CI), confirming earlier simulation results on real RNA-Seq data. No pair of classifiers differed significantly (smallest Holm-adjusted p = 0.66), and 10 of the 15 pairs not involving naive Bayes were equivalent within ±1 point. Applying calibration and conformal prediction to this dataset for the first time, to our knowledge, we found small and probably real miscalibration (softmax expected calibration error 0.47%, 95% CI 0.25–0.83, against 0.14% expected under perfect calibration),

concentrated in a few low-confidence predictions. At the 99% level, split-conformal sets gave single labels for 99.4% of samples, and abstaining on the rest lowered the error among accepted predictions from 0.51% (95% CI 0.42–0.60) to 0.18% (95% CI 0.10–0.25), similar to a plain probability threshold (0.26%); conformal prediction mainly supplies a calibrated threshold with a coverage guarantee. Class-conditional sets at 99% were uninformative because no class had the 99 calibration samples required. Linear models reached 99.4% (±0.2) macro-F1 with 10 training tumours per class. The dataset is saturated: it cannot rank modern classifiers, and on it leakage hides behind the ceiling rather than producing it. No biological interpretation is made.

Keywords: pan-cancer classification; RNA-Seq; data leakage; selection bias; equivalence testing; feature-selection stability; conformal prediction; calibration; learning curves; benchmark saturation; TCGA

1 Introduction

Identifying the tissue of origin of a tumour guides treatment, and it is most difficult in cancers of unknown primary site, which account for a few per cent of cancer diagnoses [1]. Molecular profiles carry a strong tissue-of-origin signal: the TCGA Pan-Cancer project [2] showed that cell-of-origin patterns dominate the molecular classification of 33 tumour types [3], and expression-based classifiers have been developed for clinical use [4], although a randomised trial of site-specific treatment guided by expression profiling did not improve survival [5].

A widely used small benchmark for this task is the 'gene expression cancer RNA-Seq' dataset of the UCI Machine Learning Repository [6]: 801 TCGA tumours from five cancer types and 20,531 genes. Published accuracies on it are 98–100% [7–10], and several studies fit gene selection, oversampling or principal components on the full dataset before splitting or cross-validating, use a single split, or declare a best model without a statistical test (Section 2.2). Selection on the full data is a classic source of optimistic bias in small-sample expression

studies [11–13], and single splits cannot separate classifiers whose accuracies differ by a few samples [14,15]. It is not known, however, how large the selection bias actually is on this dataset, or what conclusions about models its accuracies can support.

Research questions. (RQ1) How much does fitting selection on the full data change estimated accuracy, with real and permuted labels, at full and small sample size? (RQ2) Are seven standard classifiers distinguishable, or demonstrably equivalent? (RQ3) How many genes are needed, how stable is the selection, and what are the remaining errors? (RQ4) What can the dataset say about calibration, conformal prediction and abstention, and about how performance and leakage bias depend on sample size?

Contributions. (i) A protocol audit of published results on this dataset (Table 1). (ii) A quantified leakage audit at n = 801 and n = 60 with real and permuted labels. (iii) A repeated cross-validation benchmark of seven classifiers and a majority baseline, with all 21 pairwise corrected tests, equivalence (TOST) tests, Bayesian correlated t-tests and minimum detectable differences. (iv) Feature-count ablation, selection stability (Jaccard, Nogueira Φ) and a quality-control analysis of the persistent errors. (v) A reliability analysis applied to this dataset for the first time, to our knowledge: calibration with a perfectly calibrated reference, marginal and class-conditional conformal prediction sets nested inside the cross-validation, compared with a plain probability threshold, and selective abstention, including whether the atypical samples are flagged. (vi) Learning curves and the growth of leakage bias as the dataset shrinks, confirming known results [16,17] on this dataset with confidence intervals. (vii) Code that regenerates every number and figure (Supplementary File S1). We do not propose a new classifier.

2 Related Work

2.1 Expression-based tumour classification

Expression-based cancer classification began with microarrays [18–20]; TCGA RNA-Seq enabled pan-cancer classifiers over 31–33 tumour types with about 95–96% accuracy [21–23] and whole-transcriptome models evaluated on metastatic cancers [4]. The five-class subset studied here, with well-separated tissues, is much easier.

2.2 Published protocols on the UCI dataset

Table 1. Evaluation protocols of published studies on the UCI TCGA pan-cancer dataset (full texts read)

Study	Validation	Where selection / reduction is fitted	Reported accuracy, %	Test between models
Nitta et al. 2021 [7]	5-fold CV	SMOTE (801 → 1,500) and univariate selection on full data, before CV	98–100 (8 models)	None
Wahid & Banday 2023 [8]	Single 70:30 split	PCA (5 components) on full data, before split	99.59 (voting ensemble)	None
Alanazi et al. 2024 [9]	600/201 hold-out; 5-fold CV for tuning	ANOVA ranking and PCA; position relative to split not stated	99.995 (as reported)	None between models
Akter et al. 2025 [10]	70:30 split and 5-fold CV	Lasso/ridge/RF selection; position relative to CV not stated	99.87 (SVM, CV)	None between models
This study	10×5 repeated stratified CV	Inside every training fold; leaky variant audited	98.98–99.81 (7 models)	All 21 pairs, TOST, Bayesian

Accuracies as reported. On 201 test samples, 99.995% is not attainable (nearest: 99.50%, 100%); presumably an average or a different quantity.

Of the studies whose full texts we could read (Table 1), two fitted oversampling with selection, or PCA, on all samples before validation, two did not state where selection was fitted, and none tested differences between models.

The closest recent work is Sirohi [24], who used a larger, gene-symbol-labelled TCGA extract of the same five tumour types (2,928 tumours), recomputed random-forest importance inside every fold and found a 75-gene panel unstable (mean pairwise Jaccard 0.319) although accuracy stayed above 99% and an external cohort reached 97.7%. Our study is complementary: it addresses the anonymised UCI release that most published results use, quantifies their exposure to leakage, and tests differences and equivalence between classifiers.

2.3 Selection bias, model comparison, reliability and data efficiency

Ambroise and McLachlan showed, on microarray datasets with tens of samples, that selecting genes on the full dataset and then cross-validating gives nearly zero error even on random labels [11]; Simon et al. [12] and Varma and Simon [25] described related biases, and Kapoor and Narayanan list pre-processing on the full data among the commonest leakage types across science [13]. Vabalas et al. showed by simulation that this bias is largest at small n, persists at n = 1,000 and is smaller when the classes are easy to discriminate [17]. The stability of selected gene sets is a separate quality criterion [26]. For two classifiers on one dataset, resampled t-tests are anti-conservative because training sets overlap; Nadeau and Bengio's corrected variance addresses this [14,15], and Benavoli et al. proposed a Bayesian correlated t-test with a region of practical equivalence [27]. A non-significant difference is not evidence of equivalence; that requires an equivalence test such as two one-sided tests (TOST) [28]. Reporting guidance asks for complete, reproducible descriptions [29].

Calibration measures whether predicted probabilities can be trusted [30]; with few errors, the expected calibration error (ECE) has a non-zero sampling floor, which consistency resampling makes explicit [31]. Conformal prediction turns a classifier into one that outputs label sets with guaranteed marginal coverage under exchangeability; Mondrian variants guarantee coverage within each class, cross-conformal variants use all training data for calibration, and randomised scores handle ties [32–35]. It has been applied to expression-based classifiers for leukaemia subtyping [36], TCGA breast-cancer subtypes with a nested scheme motivated by small classes [37] and lymphoma subtypes [38]. Learning curves for expression classifiers go back to Mukherjee et al. [16]; Figueroa et al. fitted inverse power laws to predict sample-size needs [39], and Qi et al. applied this to bulk RNA-Seq [40]. In a search in October 2026 we found no calibration, conformal or learning-curve study on the UCI dataset studied here; this is absence of evidence in the sources we could find, not proof.

Small gene panels, by contrast, have been reported for it repeatedly [41,42] and their stability examined [24], so we did not pursue a minimal panel.

3 Materials and Methods

3.1 Data

We used the UCI dataset 401 (doi:10.24432/C5R88H) [6], described as a random extraction of TCGA PANCAN RNA-Seq (Illumina HiSeq) profiles [2]. It contains 801 samples (BRCA 300, KIRC 146, LUAD 141, PRAD 136, COAD 78) and 20,531 genes with no missing values. Genes are named gene_0 to gene_20530; the repository states that these are dummy names ordered as in the original Synapse submission, and no gene symbols are distributed. Values range from 0 to 20.78 and appear to be already log-transformed, so no further transformation was applied; 267 genes are constant across all samples. The SHA-256 checksums of the files are listed in the code README. A principal component analysis (Supplementary Section S1) already separates the five classes largely in two dimensions.

3.2 In-fold pipeline and classifiers

Each model was wrapped in a scikit-learn pipeline [43] refitted on the training part of every fold: (1) removal of genes with zero variance in the training fold; (2) one-way ANOVA F-test and retention of the top k genes (k = 100 in the main benchmark, fixed in advance); (3) z-scoring with training-fold statistics; (4) the classifier. Hyperparameters were fixed in advance at common defaults and not tuned, so no nested selection was needed [25]. We compared a majority-class baseline; multinomial (softmax) logistic regression with L2 penalty (C = 1); a ridge classifier (α = 10); a linear support vector machine [20] (C = 1, one-vs-rest); Gaussian naive Bayes; a random forest [44] (500 trees); LightGBM [45] (200 trees, learning rate 0.05, 15 leaves); and a multilayer perceptron with one hidden layer of 64 ReLU units (L2 penalty 10^{-3}, Adam, up to 1,000 epochs).

3.3 Leakage audit

The 'leaky' variant fitted the zero-variance filter, ANOVA selection and scaling once on all samples and then cross-validated only the classifier, on the same folds as the in-fold pipeline. (a) Real labels, n = 801: softmax, linear SVM and naive Bayes, $k \in \{10, 20, 100, 1{,}000\}$, 10×5 CV, corrected paired t-tests. (b) Permuted labels, n = 801: 100 permutations, softmax, $k \in \{10, 100\}$. (c) 50 balanced subsamples of n = 60 with real and permuted labels, mimicking the original selection-bias studies [11]. Under permutation, accuracy expected by chance is 24.3% at n = 801 (class-frequency guessing; majority rate 37.5%) and 20% at n = 60.

3.4 Evaluation and statistics

The main benchmark used 10×5 repeated stratified cross-validation (50 splits, identical for all models; seed 2026) with accuracy, macro-F1 (primary), balanced accuracy and macro ROC-AUC, and 95% intervals from the Nadeau–Bengio corrected variance [15]. All 21 pairs of classifiers were compared by the corrected paired t-test (Holm-adjusted), TOST with a ±1-point equivalence margin [28], the Bayesian correlated t-test with a ±0.5-point region of practical equivalence (ROPE) [27], and the minimum difference detectable with 80% power. Because fold scores are bounded at 100% and many folds are perfect, these t-based results are approximate.

3.5 Ablation, stability and error analysis

For four classifiers we varied k from 2 to all expressed genes (3×5 CV), and all seven were run with all expressed genes under the main protocol. Selection stability was measured by the mean pairwise Jaccard index and the Nogueira estimator Φ with a bootstrap interval [26]. Each sample received three within-class quality z-scores (mean expression, fraction of zeros, correlation with the leave-one-out class centroid), and a logistic regression on nine global

summary statistics per sample tested whether such properties differ between classes. Computational footprint is reported in Supplementary Section S9.

3.6 Reliability: calibration, conformal prediction and abstention

For the five probabilistic classifiers, calibration was assessed on the 10×801 pooled out-of-fold predictions: top-label ECE (15 bins) with a bootstrap interval over tumours, equal-mass ECE, log-loss and Brier score [30], compared with the ECE of a perfectly calibrated model with the same confidences, obtained by redrawing labels from the predicted probabilities (consistency resampling [31]). For conformal prediction [32,33], each outer training fold was split (stratified 75:25) into a proper training part, on which the whole pipeline was fitted, and a calibration part of about 160 tumours; point predictions therefore come from the 75% model. The score was one minus the probability of the label; marginal sets used one quantile, Mondrian sets one per class, which exists only with $n \geq 1/\alpha - 1$ calibration samples. A prediction was accepted only if its set was a singleton. We report coverage with its spread over folds and the calibration-conditional 5th percentile, Beta(r, n + 1 − r) with $r = \lceil (n + 1)(1 - \alpha) \rceil$ [33], and three checks: a maximum-probability threshold (same threshold, or one matching the abstention rate on calibration data); cross-conformal sets from five inner folds [34]; and randomised p-values that break score ties [35]. Accepted-error rates have 95% bootstrap intervals over repeats.

3.7 Data efficiency and leakage as a function of sample size

In each outer training fold we drew m = 3–60 tumours per class, fitted the in-fold pipeline (k = 100) with softmax, the linear SVM and naive Bayes, and evaluated on the full held-out fold (95% intervals over the 10 repeats). For leakage, we drew 30 balanced subsamples each of n = 30–390 (6–78 per class) and repeated the comparison of Section 3.3 (softmax, $k \in \{10, 100\}$) with real and permuted labels (95% intervals over subsamples).

3.8 Software and computational environment

All analyses were run in Python 3.13 with scikit-learn 1.9.1 [43], NumPy 2.5.4, SciPy 1.18.1, pandas 3.0.6 and LightGBM 4.7.0 on a Linux virtual machine (8 virtual CPUs, 16 GB RAM, no GPU). A single script (run_all.py, Supplementary File S1) regenerates every reported number, table and figure from the public data with fixed random seeds; the full run needs about 8 GB of RAM.

4 Results

4.1 Overall performance

Table 2. In-fold performance, 10×5 repeated stratified CV, top-100 ANOVA genes

Model	Accuracy, % [95% CI]	Macro-F1, % [95% CI]	ROC-AUC	Errors per repeat (range)	Perfect folds, %
Majority baseline	37.45 [37.36, 37.55]	10.90 [10.88, 10.92]	0.50000	501.0 (501–501)	—
Softmax (multinomial LR)	99.48 [98.84, 100.00]	99.48 [98.84, 100.00]	0.99999	4.2 (3–5)	48
Ridge classifier	99.68 [99.18, 100.00]	99.73 [99.30, 100.00]	0.99999	2.6 (2–3)	62
Linear SVM	99.81 [99.41, 100.00]	99.81 [99.39, 100.00]	0.99999	1.5 (0–4)	76
Gaussian naive Bayes	98.98 [98.26, 99.69]	98.90 [98.14, 99.66]	0.99440	8.2 (7–10)	16
Random forest	99.55 [98.99, 100.00]	99.53 [98.92, 100.00]	0.99999	3.6 (3–4)	52
LightGBM	99.44 [98.80, 100.00]	99.40 [98.67, 100.00]	0.99995	4.5 (3–6)	42
MLP (64 hidden units)	99.46 [98.82, 100.00]	99.44 [98.77, 100.00]	0.99999	4.3 (3–5)	46

Mean over 50 folds; 95% CI from the Nadeau–Bengio corrected variance, truncated at 100%. ROC-AUC: macro one-vs-rest, mean over folds (majority baseline 0.5). Errors per repeat: misclassified samples out of 801. Perfect folds: share of the 50 test folds with no error. Balanced accuracy and Wilson intervals: Supplementary Section S2.

All seven classifiers were far above the majority baseline (37.5% accuracy) and within 0.9 percentage points of each other in macro-F1 (Table 2; Supplementary Section S2). Mean macro-F1 was highest for the linear SVM (99.81%; 15 errors in 8,010 pooled predictions) and lowest for naive Bayes (98.90%; 82 errors). The test folds contain 160–161 samples, so a single error costs about 0.6 point of accuracy; 76% of the linear SVM's folds and 48% of softmax's were perfect. Macro ROC-AUC was at least 0.9944 for every classifier, so ranking quality does not discriminate between models.

4.2 Leakage audit

Table 3. Leaky versus in-fold selection with real labels, n = 801 (accuracy, %)

k	Model	Leaky	In-fold	Difference, points [95% CI]	p
10	Softmax	95.22	92.32	+2.89 [-2.33, +8.12]	0.271
	Linear SVM	95.37	92.31	+3.06 [-2.16, +8.27]	0.245
	Gaussian NB	95.47	90.80	+4.66 [-2.49, +11.82]	0.196
100	Softmax	99.50	99.48	+0.02 [-0.27, +0.32]	0.865
	Linear SVM	99.78	99.81	-0.04 [-0.28, +0.21]	0.759
	Gaussian NB	98.98	98.98	+0.00 [-0.23, +0.23]	1.000

Leaky: zero-variance filter, ANOVA selection and scaling fitted once on all 801 samples. 10×5 repeated CV with identical folds; corrected paired t-test, unadjusted. k = 20 and 1,000 and macro-F1: Supplementary Section S6.

With real labels and all 801 samples, fitting selection on the full data barely changed the estimate (Table 3, Fig. 1a). For $k \geq 20$ the largest leaky–in-fold difference was 0.29 point (smallest p = 0.39); for softmax with 100 genes, the setting of our main benchmark, accuracy was 99.50% leaky and 99.48% in-fold. Only with 10 genes, where the selected set itself matters, was leaky selection visibly optimistic (+2.9 to +4.7 points), and even there the corrected test was not significant ($p \geq 0.20$).

With permuted labels the bias was present but small at full size: softmax reached 38.1% (leaky) vs 33.0% (in-fold) with 10 genes and 35.9% vs 25.3% with 100 genes; the leaky pipeline thus scored about at the 37.5% majority rate, the in-fold pipeline below it (Fig. 1b). In balanced subsamples of n = 60, the bias described by Ambroise and McLachlan appeared clearly: on permuted labels leaky selection reached 51.2% (10 genes) and 68.2% (100 genes) against 19.7% and 19.9% in-fold, with 20% expected by chance (Fig. 1c). With real labels at n = 60 the difference was again small (99.9% vs 99.3% with 100 genes). The in-fold pipeline's real-label accuracy exceeded all 100 full-size permutations (permutation p = 0.010, the smallest attainable with 100 permutations).

On this dataset, therefore, leakage mainly hides behind the ceiling: leaky and leak-free pipelines give the same number, so a permutation test or a small-sample check is needed to

tell them apart, and the published accuracies in Table 1 are probably not much inflated by their protocols.

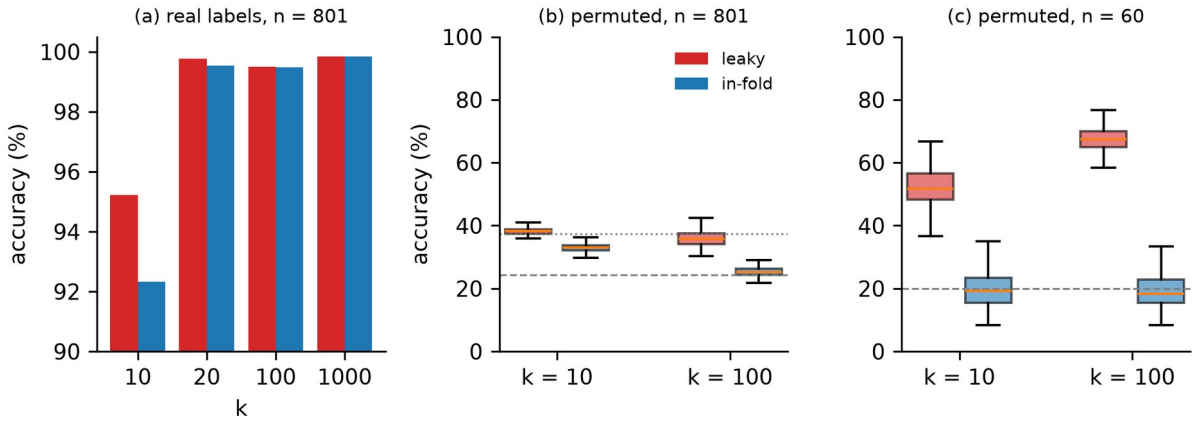


Figure 1. Leakage audit. (a) Softmax accuracy with real labels, n = 801, leaky (red) vs in-fold (blue) selection, 10×5 CV. (b) Permuted labels, n = 801, 100 permutations; dashed line: 24.3% expected by class-frequency guessing; dotted: 37.5% majority rate. (c) Permuted labels, 50 balanced subsamples of n = 60; dashed line: 20% chance.

4.3 Pairwise comparisons and equivalence

Across all 21 pairs of classifiers, no macro-F1 difference was significant after Holm adjustment; the smallest unadjusted p was 0.031 (Linear SVM vs Gaussian naive Bayes, 0.91 point), which became 0.66 after adjustment (all 21 pairs in Supplementary Section S5). Absence of a significant difference is not equivalence, so we tested equivalence directly. Within a ±1-point margin, TOST established equivalence for 10 of 21 pairs, including 10 of the 15 pairs among the six classifiers other than naive Bayes; no pair involving naive Bayes was shown equivalent. The Bayesian correlated t-test was less decisive but pointed the same way: the probability that the difference lay within ±0.5 point exceeded 0.95 for 4 pairs, and for pairs involving naive Bayes the posterior favoured the other model with probability 0.50–0.84. With 50 folds the corrected test could detect only differences of 0.5–1.4 points with 80% power, i.e. about 4–11 samples per repeat.

Differences of a few tenths of a point, one to three samples per repeat, cannot be resolved by this benchmark; where a conclusion is supported, it is equivalence.

4.4 Error analysis

Summed over the 10 repeats, softmax regression made 42 errors in 8,010 predictions (Supplementary Section S7); the most frequent confusions were LUAD predicted as BRCA (27); COAD predicted as LUAD (8); KIRC predicted as BRCA (7). All errors came from only 5 samples, each misclassified in 7–9 of the 10 repeats, and they were atypical profiles rather than random misses. The 3 LUAD samples (129, 599, 507) had low mean expression (within-class z = -2.4, -3.1, -1.9), a high fraction of zero values (z = +1.4, +2.1, +1.3) and low correlation with the LUAD centroid (z = -2.5, -3.8, -2.1); sample 599 was the least typical of all 141 LUAD tumours. The misclassified KIRC sample also had low centroid correlation (z = -3.3). Such profiles are consistent with low tumour content, degraded RNA or mislabelling, but the release contains no metadata that could distinguish these explanations.

4.5 Feature-count ablation and selection stability

Performance saturated quickly (Supplementary Section S3): all three linear models exceeded 99% macro-F1 from k = 20 genes, whereas naive Bayes peaked at 99.26% (k = 500) and fell to 56.5% with all genes. With all expressed genes under the full 10×5 protocol, ridge and the linear SVM made no error in any fold (Supplementary Section S4), so the remaining errors at k = 100 reflect the restriction to 100 genes. Across the 50 training folds, 76 genes entered the top 100 in every fold; the mean pairwise Jaccard index was 0.87 and Nogueira's Φ 0.931 (bootstrap 95% CI 0.929–0.937; Supplementary Section S8, which also lists the genes with the largest coefficients). This is much higher than the Jaccard of 0.319 that Sirohi reported for a 75-gene random-forest panel [24]; the settings differ in data, ranking method and panel size,

so stability is method-dependent rather than either estimate being wrong. Gene identities are anonymised, so we make no claim about biological roles.

4.6 Reliability: calibration, conformal prediction and abstention

Softmax had a pooled ECE of 0.47% (95% CI 0.25–0.83) against 0.14% (0.08–0.22) for a perfectly calibrated model, equal-mass ECE 0.22% and log-loss 0.0098; LightGBM (0.45%), the MLP (0.35%) and naive Bayes (1.02%; log-loss 0.34 from saturated probabilities) were also above their references, and the random forest was clearly under-confident (2.72%; Supplementary Section S11). The miscalibration lies in the few low-confidence predictions: the 80 softmax predictions with confidence below 0.93 held 42 of its 42 errors and were over-confident (Fig. 2a). As each tumour contributes 10 correlated predictions, the reference interval is somewhat too narrow; we read this as small and probably real miscalibration.

Table 4. Split-conformal prediction sets for softmax regression (k = 100), nested in 10×5 CV

Sets	Nominal, %	Coverage, %	Fold range, %	Beta 5th pct, %	LUAD cov., %	Mean size	Singleton, %	Empty, %	≥ 2 labels, %
Marginal	90	89.6	78.1–95.6	86.0	83.0	0.90	89.6	10.4	0.0
	95	94.7	85.0–99.4	92.0	89.9	0.95	94.7	5.3	0.0
	99	99.4	96.9–100.0	98.2	97.9	1.00	99.4	0.5	0.2
Mondrian	90	91.1	—	—	92.1	0.91	91.1	8.9	0.0
	95	96.3	—	—	96.4	1.87	13.4	0.0	86.6
	99	100.0	—	—	100.0	5.00	0.0	0.0	100.0

Calibration on 25% of each outer training fold (about 160 tumours: about 60 BRCA, 29 KIRC, 28 LUAD, 27 PRAD and 16 COAD); evaluation on the outer test fold. Beta 5th pct: 5th percentile of coverage conditional on the calibration set. Mondrian sets need $\geq 1/\alpha - 1$ calibration samples per class (19 at 95%, 99 at 99%), so COAD is always included at 95% and every class at 99%. Threshold baseline, cross-conformal and other models: Supplementary Section S11.

Marginal sets were valid on average (Table 4, Fig. 2b; softmax 89.6%, 94.7% and 99.4% at 90%, 95% and 99%), but fold coverage ranged from 78.1% to 95.6% at 90% and from 96.9% to 100% at 99%, wider than the calibration-conditional 5th percentiles of 86.0% and 98.2% (9 and 9 of 50 folds fall below them), because each test fold of about 160 tumours adds binomial

noise. At 99% the quantile is the largest calibration score, so the implied probability threshold ranged from 0.05 to 0.97 across folds. Naive Bayes over-covered (98.6% at 90% and 98.6% at 95%) because saturated probabilities tie the scores; randomised p-values gave 90.2%, 94.5% and 99.0%. At 90% and 95% coverage was reached through uninformative empty sets (10.4% and 5.3%), and LUAD was under-covered (83.0% at 90%).

At 99%, 99.4% of samples received one label, and accepting only these lowered the error among accepted predictions from 0.51% (95% CI 0.42–0.60) to 0.18% (95% CI 0.10–0.25) (paired difference -0.34 point, 95% CI -0.47 to -0.21). A maximum-probability rule at the same threshold accepted 13 more predictions with 7 more errors (0.26% (95% CI 0.16–0.35)), and a threshold matched on calibration data gave 0.29% (95% CI 0.16–0.43); the intervals overlap. Conformal prediction thus does not select errors better than a threshold here; it chooses the threshold with a coverage guarantee. Cross-conformal sets covered 99.1% with 99.2% singletons and 2 errors among 7,942 accepted predictions (vs 0.62% without abstention), promising but with a weaker formal guarantee ($1 - 2\alpha$). Mondrian sets fixed LUAD coverage at 90% (92.1%) but were uninformative at 95% (mean size 1.87) and 99% (all five labels): the calibration part holds about 60 BRCA, 29 KIRC, 28 LUAD, 27 PRAD and 16 COAD tumours, fewer than the 19 needed at 95% for COAD and the 99 needed at 99% for every class, the small-class problem that motivated nested designs in TCGA [37]. The three atypical LUAD samples (129, 507, 599) were flagged in 18 of their 30 test appearances (41–77%, Clopper–Pearson; 13 empty, 5 BRCA+LUAD sets) and confidently wrong 11 times; other models flagged them 12–26 times.

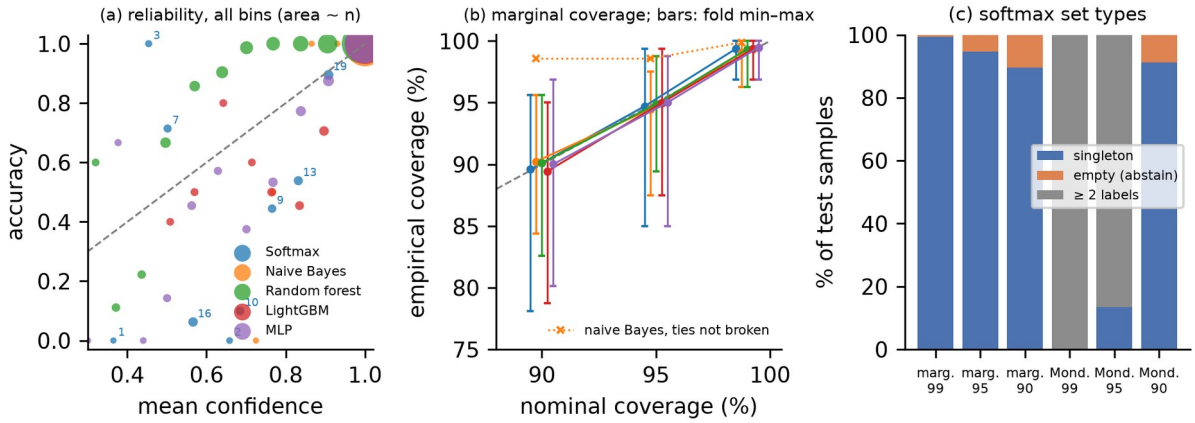


Figure 2. Calibration and conformal prediction sets. (a) Reliability diagram, 10×801 pooled out-of-fold predictions, all bins (marker area proportional to the number of predictions; softmax bin counts annotated). (b) Empirical vs nominal coverage of marginal split-conformal sets, mean and range over the 50 folds (naive Bayes with randomised p-values; crosses: without). (c) Shares of singleton, empty and multi-label sets for softmax (marg.: marginal; Mond.: Mondrian; number: nominal coverage, %).

4.7 Data efficiency and leakage as a function of sample size

The task needs very few training tumours (Fig. 3a; Supplementary Section S12): softmax reached 98.1% (±0.7) macro-F1 with 3 per class, 99.4% (±0.2) with 10 and 99.6% (±0.1) with 60, and the linear SVM 98.3% (±0.7), 99.4% (±0.2) and 99.8% (±0.04). Naive Bayes needed far more data (74.0% (±2.4) at 10, 97.9% (±0.3) at 60) and was non-monotone at the smallest sizes. As predicted by Vabalas et al. [17], leakage bias on permuted labels shrank with n but did not vanish (Fig. 3b): with 100 genes, 65.1 ± 4.8, 49.2 ± 4.7, 35.6 ± 3.5, 21.0 ± 1.8 and 14.1 ± 1.7 points at n = 30, 60, 120, 240 and 390, and 10.6 ± 0.5 points in the separate, unbalanced n = 801 design. With real labels and 100 genes it never exceeded 1.0 ± 0.7 points; with 10 genes it was 3.7 ± 2.0 at n = 60 and turned negative at n = 390 (-1.6 ± 0.8), i.e. full-data selection made the estimate slightly worse. This confirms, on real RNA-Seq data, a known mechanism: when a few tumours per class reach the ceiling, selection bias has no room to raise the estimate.

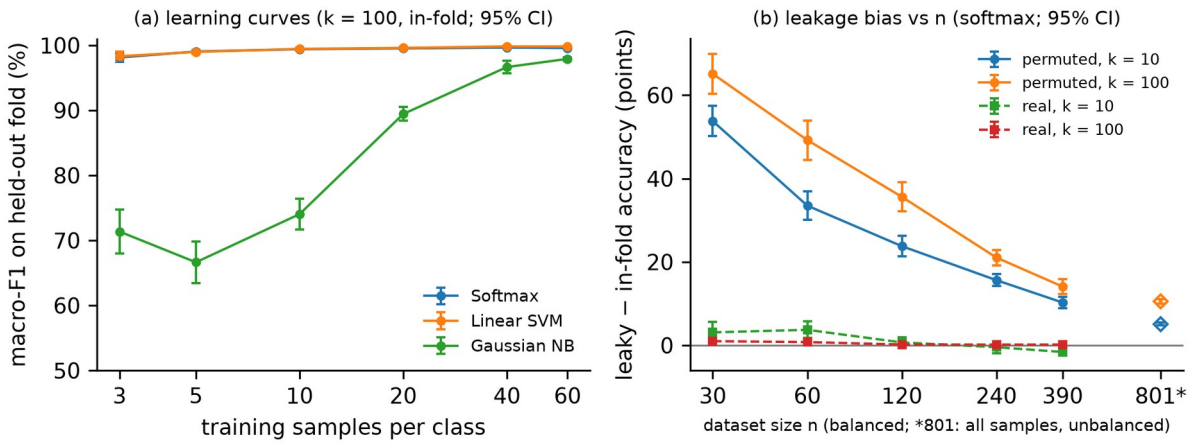


Figure 3. Data efficiency and leakage bias by sample size. (a) Macro-F1 on the full held-out fold against the number of training tumours per class (balanced draws from each of the 50 training folds; mean and 95% CI over 10 repeats). (b) Leaky minus in-fold accuracy of softmax in 30 balanced subsamples per n (mean and 95% CI), with permuted (solid) and real (dashed) labels; open diamonds: all 801 samples, unbalanced classes, 100 permutations (different design).

5 Discussion

The original motivation for this kind of study, that leaky selection inflates published accuracies, turns out to be weak on this dataset at its full size: with real labels, leaky and in-fold selection gave practically identical accuracy, and the bias of Ambroise and McLachlan [11] appeared clearly only with random labels in small samples, as the simulations of Vabalas et al. predict for easily separable classes [17]. The tissue-of-origin signal is so strong [3] that every reasonable pipeline reaches the ceiling. Published results are therefore probably numerically sound despite their protocols, but the dataset cannot demonstrate that a pipeline is free of leakage, nor rank classifiers: the smallest detectable difference was 0.5–1.4 points, larger than the differences among the six non-naive-Bayes classifiers, 10 of whose 15 pairs were equivalent within ±1 point. Selection stability is likewise a property of the whole pipeline: high here ($\Phi = 0.93$), low for a small random-forest panel elsewhere [24].

The reliability analyses show what a saturated benchmark can still contribute, and their limits. Miscalibration was small and confined to a few low-confidence predictions, and 99% conformal sets flagged about half of the occurrences of the atypical samples while leaving 99.4% of predictions as single labels. Conformal sets did not, however, select errors better than a probability threshold; their contribution is a threshold chosen with a coverage guarantee, and with about 160 calibration tumours that threshold varies considerably between calibration sets. At 90% and 95% the sets were uninformative for a classifier this accurate, and class-conditional guarantees at 99% need more calibration samples per class than the dataset provides. These findings are specific to this dataset and to within-TCGA evaluation, not evidence about clinical performance.

We recommend that studies using this dataset fit all selection, resampling and scaling inside the training folds and say so; report repeated cross-validation with corrected intervals, all pairwise tests with multiplicity control and an equivalence test when claiming that models perform alike; treat a permutation test, not the accuracy level, as evidence of a leak-free pipeline; avoid biological claims about anonymised genes; and, if reliability is claimed, compare calibration with its sampling floor, report conformal coverage per class and across calibration sets, and compare abstention with a plain probability threshold.

6 Conclusion

On the UCI TCGA pan-cancer RNA-Seq dataset, near-perfect accuracy is the norm under both leaky and leak-free protocols. At full sample size leaky selection barely changes the estimate, so the dataset neither exposes leakage nor ranks modern classifiers; no pair of seven standard classifiers differs significantly, 10 of the 15 pairs among the six non-naive-Bayes models are demonstrably equivalent within ±1 percentage point, and naive Bayes is probably slightly worse. Its value is as a teaching and sanity-check example, with permutation tests and equivalence testing, and as a test bed for reliability: 99% conformal sets allow a small amount

of abstention to remove most remaining errors, much as a probability threshold would, while class-conditional guarantees need more data. Claims of methodological progress require harder, many-class and cross-cohort settings.

7 Limitations and Future Work

The dataset has 801 samples from one source (TCGA), five well-separated tissue types and no clinical metadata. Global expression summaries alone predicted the cancer type with 52.7% balanced accuracy, so classes also differ in overall profile properties (biology, handling or batch [46]). Cross-validation estimates generalisation within TCGA, not to other cohorts or metastatic samples [1,4]. Hyperparameters were fixed. The t-based tests assume approximately normal fold differences, which bounded scores violate; the conclusions rest on several analyses agreeing. The leakage audit covers ANOVA selection and scaling, not oversampling or PCA. Conformal guarantees assume exchangeability; with about 160 calibration tumours, coverage varies widely between calibration sets, and the cross-conformal guarantee is weaker. Calibration references treat the ten predictions per tumour as independent. The learning curves use one balanced draw per fold and one k, without fitted learning-curve models; leakage-versus-n subsamples overlap heavily at n = 390 (all 78 COAD tumours each time), and the n = 801 permutations use a different, unbalanced design. Mapping the anonymised genes to symbols via the original Synapse annotation, external validation and the 33-type problem are left for future work.

Data Availability Statement

The dataset is publicly available from the UCI Machine Learning Repository (doi:10.24432/C5R88H) [6]. All code, per-fold results and figures are provided in Supplementary File S1 and at [Repository URL / DOI]. Code licence: [MIT or other licence]. Software: PanCancer_code_S1 (version as submitted; archived version: [DOI to be added]);

language Python 3.13; platform Linux (any operating system with Python 3.13 should work); dependencies listed in requirements.txt (scikit-learn, NumPy, SciPy, pandas, LightGBM, matplotlib).

Ethics Statement

This study analysed only publicly available, de-identified gene-expression data released for research use; no new data were collected. Ethics approval and consent to participate were therefore not required for this secondary analysis. Consent for publication: not applicable.

Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process

[Placeholder to finalise.] AI agents were used to write the analysis code, run the experiments and draft and edit the manuscript text. All content was reviewed, verified and edited before submission, and responsibility for the publication rests with the named author(s). All numerical results were produced by the released code.

Declaration of Conflict of Interest

No conflict of interest is declared. [To confirm.]

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[To confirm; add any co-authors or supervisor and their roles.] Rishabh Aryan: conceptualisation, study design, supervision of the AI-assisted analysis, verification of results, and review and approval of the final manuscript.

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